

Emergent Vacuum Response in Nonequilibrium Electromagnetic Systems: A Constrained Phenomenological Framework with Experimental Pathways

Erick S. Sangalang

2026

Abstract

This work presents a consolidated framework for Emergent Vacuum Response Theory (EVRT), a phenomenological approach investigating whether structured, nonequilibrium electromagnetic systems may exhibit measurable stress-energy effects beyond conventional interpretations. Rather than asserting the existence of new physics, the framework is constructed to remain consistent with established conservation laws while introducing a constrained effective response parameter subject to experimental bounds.

The paper synthesizes prior theoretical, numerical, and experimental considerations into a unified structure emphasizing falsifiability, parameter constraints, and reproducibility. Simulation-based explorations illustrate qualitative behavior under representative conditions, while proposed experimental protocols outline strategies for distinguishing genuine signals from thermal, vibrational, electromagnetic, and measurement artifacts.

Importantly, the framework treats null results as scientifically meaningful, enabling the establishment of upper bounds on any emergent response. The goal is not to claim anomalous force generation, but to define a disciplined pathway for testing whether such effects exist and, if not observed, to constrain the parameter space in which they could occur.

1 Introduction

The interaction between electromagnetic fields and vacuum structure has long been explored within the context of classical electrodynamics and quantum field theory. While standard formulations do not predict macroscopic force asymmetries in closed electromagnetic systems, recent interest in highly asymmetric geometries and nonequilibrium configurations has motivated renewed experimental and theoretical scrutiny.

This work introduces Emergent Vacuum Response Theory (EVRT), a constrained phenomenological framework designed to investigate whether structured electromagnetic systems could exhibit small residual stress-energy effects beyond conventional expectations. The aim is not to assert the presence of new physics, but to define a falsifiable structure within which such possibilities can be rigorously tested.

2 Theoretical Framework

As shown in Figure 1, the EVRT framework follows a structured progression from theoretical formulation through experimental evaluation.

EVRT is constructed as an effective phenomenological extension consistent with classical electromagnetic theory. The framework introduces an effective response parameter, denoted χ_{eff} , representing a potential emergent coupling between structured electromagnetic fields and an effective vacuum response.

The standard electromagnetic stress-energy tensor remains the baseline description:

$$T^{\mu\nu} = \epsilon_0 \left(E^\mu E^\nu + c^2 B^\mu B^\nu - \frac{1}{2} \eta^{\mu\nu} (E^2 + c^2 B^2) \right) \quad (1)$$

3 Baseline Physical Assumptions and EVRT Extension

3.1 Baseline Physical Assumptions

The present work adopts standard classical electromagnetic theory as its baseline. In particular, the following assumptions are maintained:

- Maxwell’s equations fully describe electromagnetic field dynamics in vacuum and material media.
- The electromagnetic stress-energy tensor governs momentum and energy transport.
- Conservation of energy and momentum holds in all closed systems.
- Symmetric or closed electromagnetic configurations do not produce net external forces in the absence of external interactions.

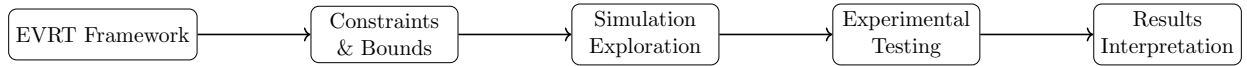


Figure 1: Conceptual pipeline of the EVRT research program. The framework progresses from phenomenological formulation through constraint analysis, simulation-based exploration, and experimental testing, culminating in interpretation via null or non-null outcomes.

These principles form the reference framework against which any proposed extension must be evaluated. Any deviation from expected behavior must be consistent with these constraints or explicitly bounded within experimental uncertainty.

3.2 EVRT Phenomenological Extension

Emergent Vacuum Response Theory (EVRT) introduces a constrained phenomenological extension to this baseline framework. The central hypothesis is that under certain structured, nonequilibrium electromagnetic conditions, an effective response of the vacuum may be parameterized by a dimensionless quantity χ_{eff} .

This parameter is defined operationally as:

- $\chi_{\text{eff}} = 0$: system behavior is fully described by standard electrodynamics.
- $\chi_{\text{eff}} \neq 0$: an additional effective contribution to the stress-energy response may be present under specific conditions.

The EVRT extension is not derived from a microscopic model but is instead constructed as a testable phenomenological parameterization subject to the following constraints:

- It must not violate global conservation laws.
- It must vanish in symmetric or equilibrium configurations.
- It must remain consistent with existing experimental null results.
- It must produce effects, if present, within experimentally accessible sensitivity ranges.

Within this framework, any observable deviation from standard predictions is interpreted as a constraint on or potential indication of a nonzero χ_{eff} , subject to rigorous exclusion of conventional physical explanations.

The EVRT framework does not imply violation of momentum conservation; any effective force interpretation must be consistent with a complete accounting of field and system momentum.

3.3 Scope of Applicability

The EVRT extension is restricted to regimes characterized by:

- strong field asymmetry or geometric anisotropy
- nonequilibrium or driven electromagnetic conditions
- resonant or high-quality-factor structures

Outside of these regimes, standard electromagnetic theory is expected to fully describe system behavior, and χ_{eff} is assumed to effectively vanish.

This separation ensures that EVRT remains a constrained, falsifiable extension rather than a general modification of established physics.

EVRT posits that under strongly nonequilibrium or asymmetric boundary conditions, an additional effective contribution may arise:

$$T_{\text{eff}}^{\mu\nu} = T^{\mu\nu} + \chi_{\text{eff}} \Delta T^{\mu\nu} \quad (2)$$

where $\Delta T^{\mu\nu}$ represents a structured perturbative term subject to conservation constraints.

4 Constraint Analysis

Any emergent contribution must remain consistent with:

- conservation of energy and momentum
- absence of net force in symmetric configurations
- experimentally established null results

Order-of-magnitude considerations suggest that if present, χ_{eff} must be small, placing predicted forces in the nanonewton or sub-nanonewton regime for laboratory-scale systems.

These constraints significantly limit the parameter space and emphasize the importance of high-precision experimental validation.

5 Simulation Overview

Numerical simulations within this framework are intended as illustrative tools rather than predictive models. Simplified representations of asymmetric resonant structures can produce qualitative behavior consistent with small residual responses under certain parameter choices.

However, these simulations do not constitute evidence of physical effects and must be interpreted cautiously. Their primary role is to guide experimental design and identify parameter regimes of interest.

6 Experimental Protocols

Testing EVRT requires careful elimination of known artifacts, including:

- thermal drift
- mechanical vibration
- electromagnetic interference
- ion wind and electrohydrodynamic effects

A baseline experimental approach involves:

- high-voltage asymmetric structures or resonant cavities
- vacuum operation where feasible
- precision force measurement (e.g., torsion balance or microbalance)
- repeated controlled trials with power cycling

7 Testable Predictions

A central goal of the EVRT framework is to produce experimentally distinguishable outcomes that can be used to either constrain or potentially detect a nonzero effective response parameter χ_{eff} . The following predictions are formulated to be directly testable under controlled laboratory conditions.

7.1 Null Hypothesis (Standard Electrodynamics)

Under the assumption that $\chi_{\text{eff}} = 0$, all observed system behavior should be fully explained by known physical mechanisms. In this case:

- No statistically significant net force should be observed in closed or symmetric electromagnetic configurations.
- Any measured force signals should be attributable to known effects, including thermal drift, mechanical vibration, electromagnetic interference, or electrohydrodynamic forces.
- Observed signals should not persist under improved experimental controls such as vacuum operation or enhanced isolation.

This null hypothesis provides the baseline against which all experimental observations must be compared.

7.2 Non-Null Hypothesis (EVRT Response)

If $\chi_{\text{eff}} \neq 0$ under specific nonequilibrium and asymmetric conditions, the following qualitative signatures may be observed:

- A small but reproducible residual force correlated with electromagnetic field asymmetry or boundary conditions.
- Persistence of the signal under vacuum conditions, reducing the likelihood of electrohydrodynamic effects.
- Dependence of the signal magnitude on input power, field strength, or resonant conditions.
- Reproducibility across repeated trials with consistent system configuration.

These signatures must be evaluated alongside rigorous artifact elimination to distinguish genuine effects from systematic error.

7.3 Scaling Behavior

If present, EVRT-induced effects are expected to exhibit scaling behavior consistent with system parameters. Qualitatively, this includes:

- Increasing response with field intensity or stored electromagnetic energy.
- Sensitivity to geometric asymmetry and boundary conditions.
- Potential enhancement under resonant or high-quality-factor conditions.

The absence of such scaling would argue against an EVRT-consistent interpretation.

7.4 Experimental Discrimination Criteria

To distinguish between null and non-null outcomes, the following criteria must be satisfied for any claimed detection:

- Signal magnitude must exceed statistical and systematic uncertainty.
- Signal must persist under control conditions designed to eliminate known artifacts.
- Independent replication must confirm the effect under comparable conditions.

Failure to meet these criteria results in classification as a null outcome, contributing to upper bounds on χ_{eff} .

7.5 Interpretation of Outcomes

The EVRT framework is constructed such that both null and non-null results are scientifically meaningful:

- Null results constrain the allowable parameter space of χ_{eff} and reinforce the validity of standard electrodynamics within tested regimes.
- Non-null results, if confirmed, would motivate further investigation into underlying mechanisms while remaining subject to rigorous verification.

This structure ensures that the framework remains falsifiable and experimentally grounded regardless of outcome.

8 Results Interpretation

Outcomes are categorized as:

Null Result: No statistically significant force beyond measurement noise. This constrains χ_{eff} and reduces viable parameter space.

Non-Null Result: Persistent, reproducible signal exceeding noise thresholds. Such a result requires rigorous artifact elimination and independent replication.

9 Discussion

Within experimental sensitivity limits typical of tabletop systems, null results at the nanonewton scale would constrain χ_{eff} to values significantly below unity, implying that any emergent response—if present—must be weakly coupled under accessible laboratory conditions.

The EVRT framework is designed to produce meaningful results regardless of outcome. Null findings refine theoretical constraints, while any verified non-null result would motivate deeper investigation into underlying mechanisms.

Limitations of the current framework include its phenomenological nature and lack of a microscopic derivation. Future work may explore connections to quantum vacuum effects or nonequilibrium field theory.

Achieving meaningful constraints on χ_{eff} requires force sensitivity at or below the nanonewton scale, placing practical limits on experimental implementation.

10 Conclusion

The framework therefore provides a structured pathway for either detecting novel effects or strengthening confidence in existing electromagnetic theory through progressively tighter experimental bounds.

As summarized in Figure 1, the EVRT framework integrates theoretical, numerical, and experimental components into a unified, testable structure.

This work presents a structured, falsifiable framework for investigating potential emergent vacuum responses in nonequilibrium electromagnetic systems. By emphasizing constraint analysis, reproducibility, and experimental rigor, EVRT provides a disciplined pathway for testing speculative extensions while remaining grounded in established physics.

Data and Code Availability

All simulation tools, data, and analysis scripts are publicly available at:

<https://github.com/HitojinKyoshi>

Citation

This work is archived on Zenodo:

DOI: <https://doi.org/10.5281/zenodo.8475>

A Order-of-Magnitude Constraints on χ_{eff}

The EVRT framework introduces an effective dimensionless parameter χ_{eff} to characterize potential deviations from standard electromagnetic stress-energy behavior under structured, nonequilibrium conditions. While no specific microscopic mechanism is assumed, it is useful to estimate the magnitude of χ_{eff} that would be consistent with experimentally accessible force sensitivities.

Consider a representative laboratory-scale electromagnetic system with characteristic electric field amplitude E and volume V . The electromagnetic energy density is approximately given by

$$u \sim \frac{1}{2} \epsilon_0 E^2. \quad (3)$$

If an effective EVRT-induced contribution produces a net force F_{EVRT} , a simple dimensional estimate can be constructed by assuming that a fraction χ_{eff} of the stored electromagnetic energy density contributes to an effective directional stress over a characteristic length scale L :

$$F_{\text{EVRT}} \sim \chi_{\text{eff}} \frac{uV}{L}. \quad (4)$$

Substituting representative values for a tabletop system,

- $E \sim 10^5 \text{ V/m}$
- $V \sim 10^{-4} \text{ m}^3$
- $L \sim 0.1 \text{ m}$

yields

$$u \sim \frac{1}{2} \epsilon_0 E^2 \sim 4 \times 10^{-2} \text{ J/m}^3, \quad (5)$$

and therefore

$$F_{\text{EVRT}} \sim \chi_{\text{eff}} \times 4 \times 10^{-5} \text{ N}. \quad (6)$$

For experimental sensitivity at the nanonewton scale ($\sim 10^{-9} \text{ N}$), this implies

$$\chi_{\text{eff}} \lesssim 10^{-4}. \quad (7)$$

Figure 2 provides a schematic interpretation of these bounds in terms of experimental sensitivity.

This estimate is highly approximate and intended only to illustrate the scale of constraints accessible to laboratory experiments. More precise bounds would depend on detailed system geometry, field distribution, and measurement sensitivity.

Importantly, this result reinforces that any observable EVRT-like effect, if present, must be weakly coupled under accessible experimental conditions. Conversely, null results at increasing sensitivity directly translate into progressively tighter bounds on χ_{eff} , further constraining the viable parameter space of the framework.

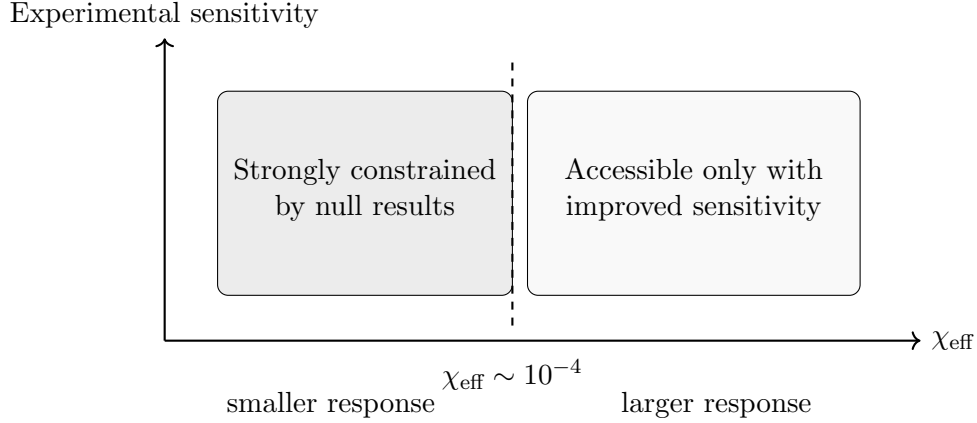


Figure 2: Conceptual parameter-space interpretation of order-of-magnitude bounds on χ_{eff} . Null results at nanonewton sensitivity constrain the effective response parameter below the illustrative bound, while improved experimental sensitivity would progressively reduce the viable parameter space. This figure is schematic and not a calibrated exclusion plot.

References

1. J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley, 1999.
2. D. J. Griffiths, *Introduction to Electrodynamics*, 4th ed., Pearson, 2013.
3. L. D. Landau and E. M. Lifshitz, *The Classical Theory of Fields*, 4th ed., Butterworth-Heinemann, 1975.
4. L. D. Landau and E. M. Lifshitz, *Electrodynamics of Continuous Media*, 2nd ed., Pergamon Press, 1984.
5. R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, Vol. II, Addison-Wesley, 1964.
6. P. W. Milonni, *The Quantum Vacuum: An Introduction to Quantum Electrodynamics*, Academic Press, 1994.
7. H. B. G. Casimir, “On the attraction between two perfectly conducting plates,” *Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen*, vol. 51, pp. 793–795, 1948.
8. W. G. Unruh, “Notes on black-hole evaporation,” *Physical Review D*, vol. 14, no. 4, pp. 870–892, 1976.
9. A. Taflov and S. C. Hagness, *Computational Electrodynamics: The Finite-Difference Time-Domain Method*, 3rd ed., Artech House, 2005.
10. K. S. Yee, “Numerical solution of initial boundary value problems involving Maxwell’s equations in isotropic media,” *IEEE Transactions on Antennas and Propagation*, vol. 14, no. 3, pp. 302–307, 1966.
11. R. D. Hazeltine and F. L. Waelbroeck, *The Framework of Plasma Physics*, Westview Press, 2004.

12. E. S. Sangalang, *Emergent Vacuum Response Theory (EVRT): Consolidated Research Framework*, Zenodo, 2026. DOI: <https://doi.org/10.5281/zenodo.8475>.